Project Specifications
– Estate Mind Project –

4DS5 – Groupe 5

Plan

1. Context & Understanding of the Problem

Real estate is a key sector in Tunisia, affecting households, investors and economic actors alike.

At the international level, this sector is undergoing a major transformation driven by PropTech platforms based on data and artificial intelligence.

In Tunisia, such approaches remain limited, which creates a real opportunity for local innovation.

It is within this context that our project takes shape: **Estate Mind**.

2. Problem Statement & Stakeholders

The Tunisian real-estate market is experiencing strong growth, with annual returns exceeding nine percent. However, despite this apparent dynamism, the process of buying, selling, or investing in property remains complex and risky for many stakeholders.

For **buyers**, the main challenges lie in navigating a large number of heterogeneous listings, assessing their credibility, avoiding overpriced properties, and identifying potential hidden defects or legal issues before committing to a purchase.

Investors face a different but equally critical set of difficulties. They must monitor emerging neighborhoods, analyze market trends and seasonal effects, anticipate future value appreciation, and minimize exposure to risky projects or unprofitable investments. The lack of consolidated and reliable market intelligence makes these strategic decisions particularly uncertain.

Sellers, on the other hand, often struggle to determine a fair and competitive price for their assets. Conflicting valuations, limited access to comparable transactions, and concerns about regulatory compliance can delay transactions and reduce market efficiency.

These difficulties are further amplified by several structural characteristics of the Tunisian market, including fragmented and heterogeneous data sources, strong regional disparities in pricing, the coexistence of transactions in dinars and foreign currencies, pronounced tourism-driven seasonality, and a complex regulatory framework related to land ownership and foreign investment.

In this context, the central question of the project can be formulated as follows:
How can we optimize real estate decisions in a fragmented and regulated market?

3. Existing Solutions & Added Value

To benchmark our project, we analyzed several international PropTech platforms.

In the **US**, Zillow provides accurate automated price estimates thanks to large public registries, but this approach depends on highly structured data — something rarely available in fragmented markets like Tunisia.

In **Europe**, platforms such as Rightmove or Idealista offer price maps and historical trends, but pay limited attention to listing verification, fraud detection or legal risk assessment, which are central issues for Tunisian buyers and investors.

In **Asia**, investor-oriented platforms like PropertyGuru focus on macro indicators and regional dashboards, but they provide little support for operational decisions on individual properties.

These examples show a clear gap: existing solutions are strong at market analytics, but **weaker when it comes to data reliability,** regulatory **risk management** and a**daptation to emerging markets**.

Estate Mind is designed precisely around these blind spots. We combine multi-source data aggregation, trust scoring, explainable valuation, territorial intelligence and a RegTech layer for legal compliance.

Unlike most platforms, we address buyers, investors, sellers and professionals at the same time, with a decision-oriented and legally aware system tailored to the Tunisian context.

4. Business Objectives & Data-Science Objectives

The Estate Mind project pursues a dual ambition: to address the concrete challenges of the Tunisian real estate market while leveraging advanced Data Science methods to produce reliable, explainable, and actionable analyses.

1	BO	Improve transparency and reliability of the Tunisian real estate market by centralizing data from multiple, fragmented, and multi-currency sources.
	DSO	1 — Automated Multi-Source Ingestion Deploy an autonomous data collection agent ensuring continuous and incremental ingestion of listings through scraping with dynamic scrolling and APIs, covering sales, land, and rental data.
		2 — Listing Structuring and Quality Control Implement NLP-based pipelines and LLMs orchestrated via LangChain to structure listings, enrich missing fields, and ensure validation and quality control of collected data, including duplicate detection, price anomaly detection, and Trust Score computation.
		3 — Data Historization, Versioning, and Orchestration Implement data historization and versioning using a Data Lake and MLflow, coupled with a pipeline orchestrator, to ensure data traceability and reproducibility.
2	BO	Understand Territorial and Temporal Dynamics of Real Estate Activity in Tunisia Understand the territorial and temporal dynamics of real estate activity in Tunisia in order to identify emerging areas and market trends.
	DSO	1 — Temporal Analysis of Real Estate Activity Model the temporal evolution of listing volumes by area using time-series methods to identify trends, seasonality, and abnormal variations in territorial activity, integrating regional macroeconomic signals when available (regional tourism evolution, inflation, exchange rates, infrastructure projects—metro, industrial zones, wind farms, mobility data—, and demographic dynamics by governorate).
		2 — Spatial Analysis and Territorial Mapping Build spatial aggregation models and geographic representations to project listings onto maps, analyze neighborhood-level density, and characterize the territorial structure of the real estate market.
		3 — Emerging Area Detection and Alert Generation Develop analytical mechanisms to automatically identify fast-growing areas, detect significant territorial changes, and trigger alerts and notifications for users.

3	BO	Explainable and Reliable Property Price Estimation. Provide a reliable and explainable estimation of real estate prices by taking into account territorial, seasonal, and macroeconomic contexts.
	DSO	1 — Predictive Price Modeling Build supervised models to estimate property values by incorporating temporal, territorial, seasonal, and macroeconomic dynamics.
		2 — Interpretability and Transparency of Price Estimates Deploy explainable AI (XAI) methods to justify price predictions, identify key driving factors, and strengthen user trust.
		3 — Anomaly Detection and Over/Under-Valuation Develop analytical mechanisms to identify overvalued or undervalued properties and automatically flag atypical pricing situations.
		4 — Integration of Macro-Territorial Signals Integrate exogenous variables into the models, such as demographic dynamics by governorate, infrastructure projects, regional tourism evolution, exchange rates, inflation, and mortgage credit indicators, in order to anticipate market trends.
		5 — Multi-Modal Valuation System Design a valuation approach combining physical property characteristics, listing textual content, and urban environment features to better capture the true quality and attractiveness of properties.
4	BO	Optimize real estate decisions for buyers and investors by providing personalized analyses, profitability simulations, and recommendations tailored to user profiles.
	DSO	1 — User Profiling & Personalization Build user profiles (personas) based on preferences and behaviors in order to tailor real estate analyses, recommendations, and strategies to individual goals.
		2 — Profitability Simulation & Decision Scenarios Develop analytical models to estimate real estate investment returns, simulate different “what-if” scenarios, and compare buy, sell, or wait strategies.
		3 — Recommendation Engine & Advice Generation Implement a recommendation engine capable of suggesting relevant properties, ensuring geographic diversity, generating targeted alerts, and producing personalized decision summaries.

5	BO	Reduce legal and regulatory risks related to real estate transactions by automating compliance analysis, enabling early detection of non-compliance, and continuously monitoring regulatory changes.
	DSO	1 — Regulatory Modeling Using a Knowledge Graph Structure legal and urban planning rules into a regulatory knowledge graph linking territories, property types, and compliance statuses, providing a usable foundation for automated analysis.
		2 — Compliance Risk Assessment & Early Detection Develop a legal agent capable of analyzing listing information, querying the regulatory knowledge graph, and producing a Regulatory Risk Score along with indicators of potential non-compliance.
		3 — Multi-Agent Orchestration & Alerts Implement a multi-agent orchestration system (legal, socio-economic) to continuously update regulations, monitor the market, and trigger real-time alerts and notifications.
		4 — Intelligent Legal Assistant Based on RAG Develop a regulatory assistant using a Retrieval-Augmented Generation (RAG) approach on accessible legal texts (Real Estate Law, case law, regulations) to provide contextualized answers based on property location and type.
		5 — Transaction Cost Simulation Design models to estimate taxes, duties, and administrative fees related to real estate transactions, and integrate these costs into investment scenarios and decision-support tools.
6	BO	Making the platform accessible and operational through a conversational assistant, automated reports, and interpretable explanations.
	DSO	1 — Intelligent Interaction with the Platform Enable users to query data, models, and analytical results through a conversational assistant connected to the analytical modules.
		2 — Automated Report and Summary Generation Generate NLP-based listing summaries and actionable PDF reports derived from analytical outputs.
		3 — Explainability and Traceability of Reasoning Provide interpretable, multi-modal explanations and maintain a history of generated responses for traceability and auditability.

5. KPIs

BO 1	Technical KPIs	1- Source coverage rate (%) 2- Data collection success rate 3- Automatically structured listings rate (%)
	Business KPIs	1- Reliable listings rate after quality control (%) 2- Duplicate reduction across platforms (%) 3- Geographic coverage of the Tunisian market (%)
BO 2	Technical KPIs	1- Correctly identified emerging zones rate 2- Correctly identified emerging zones rate 3- Detection delay of significant territorial changes
	Business KPIs	1- Number of emerging zones identified and monitored 2- Usage rate of territorial maps 3- Adoption of territorial insights by users
BO 3	Technical KPIs	1- Average price estimation error (MAE / RMSE) 2- Temporal stability of model performance 3-Quality of model explanations (XAI)
	Business KPIs	1- Average gap between asking price and estimated price 2- User adoption of the price estimation tool 3- Perceived trust in price estimations
BO 4	Technical KPIs	1- Recommendation engine performance 2- Investment return simulation accuracy 3- Relevant opportunity alert rate
	Business KPIs	1- Average investment return improvement 2- Decision time reduction 3- Adoption rate of recommendations
BO 5	Technical KPIs	1- Regulatory risk score accuracy 2- False positive / false negative rate of legal alerts 3- Compliance evaluation response time
	Business KPIs	1- Reduction of risky transaction 2- Early detection of non-compliant listings 3- Perceived trust in legal compliance
BO 6	Technical KPIs	1- Relevant response rate 2- Error / hallucination rate 3- Assistant availability (% uptime)
	Business KPIs	1- Assistant adoption rate 2- Manual analysis time reduction 3- User satisfaction with answer clarity

6. TDSP

The Estate Mind project will follow the Team Data Science Process (TDSP) methodology, a structured and iterative framework guiding data-science projects from business needs to operational deployment.

This methodology is organized into four main phases:

- Business Understanding – framing the problem, stakeholders, objectives, and success criteria.
- Data Acquisition & Understanding – collecting, exploring, and assessing data quality.
- Modeling – building, training, and evaluating predictive and analytical models.
- Deployment – integrating models into applications and ensuring scalability, monitoring, and security.

Based on the TDSP framework, roles within the Estate Mind team have been assigned to ensure effective collaboration across the four phases of the project and clear responsibility for each major activity.

Role	Main Responsibilities	Member
Project Manager	Planning, coordination, management of time, resources, and risks	BAHAR Wissem
Project Lead	Product vision, technical decision-making	BAHAR Wissal
Data Scientist(s)	Modeling, XAI, forecasting, data analysis	ALAYA Salma MRAIEH Nourhen
Solution Architect (s)	Architecture, API, deployment, pipelines	BEN MAHMOUD Yosser
Additional Roles	Main Responsibilities	Member
Business Analyst	Use cases, KPIs, regulatory compliance	ZAGHOUANI Manar
Data Engineer	Scraping, pipelines, data quality	

7. SDG

Estate Mind promotes transparency, secure transactions, and data-driven urban planning in the Tunisian real estate market, with positive economic, social, and territorial impact.

Key Contributions:

SDG 11: Responsible urbanization (price mapping, emerging areas, planning rules)

SDG 16: Trust in transactions (fraud detection, compliance alerts, legal assistance)

SDG 12: Responsible decisions (explainable estimates, comparisons, investment simulations)

Additional: SDG 9 (digital modernization), SDG 10 (territorial equity), SDG 8 (secure investments)